AAYUSH DAYAL

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EDUCATION

University of British Columbia — September 2021 – *April 2026 (Anticipated)*
Bachelor of Applied Science – Mechanical Engineering

PROFESSIONAL EXPERIENCE

TELIX – ARTMS INC. — **Kelowna, BC**
MECHANICAL ENGINEERING CAPSTONE STUDENT — ***September 2025 – April 2026***
Target Feedback & Detection System for ARTMS QUANTM Irradiation System (QIS)

- Designed capsule tracking sensor system, achieving ±0.5mm positional accuracy through modular coupling.
- Developed custom coupling modules with ToF sensors, accelerometers, & inductive PCBs for capsule detection.
- Fabricated pneumatic transfer assembly in 4-10 PSI environments to verify validation for detection & response time.
- Engineered PCB with CAN-bus protocol connecting checkpoint sensors to central hub for real-time HMI integration.
- Assisted Siemens TIA Portal dev. with capsule tracking visualization, alarm configuration, & PLC tags for operator.

VICE-PRINCIPAL RESEARCH & INNOVATION OFFICE - UBC — **Kelowna, BC**
PROJECT MANAGER — ***May 2025 – February 2026***

- Liaised with cross-functional stakeholders across research teams to align project deliverables & requirements.
- Conduct space audits & taxonomical equipment classification for 1000+ capital assets using Excel & MS Access.
- Engineered Access database with ERD design, SQL queries, VBA programming, & automated validation controls.
- Create technical documents, status reports, risk assessments supporting multi-disciplinary research facility planning

RANOSON SPRINGS PVT. LTD — **New Delhi, India**
CAD INTERN — ***May 2023 – August 2023***

- Collaborated on 3D models/assemblies on SolidWorks & maintained ASME Y14.5 standards for timely completion.
- Optimized tooling design & automated conveyor system using ISO 9001 procedures, expediting production by 15%.
- Performed quality control validation using GD&T, ensuring compliance with client contractual & spec. requirements.
- Updated 25+ legacy blueprints with revised specification, eliminating 90% of technical CAD drawing discrepancies.

TECHNICAL PROJECTS

OK MOTORRAD MotoStudent Design Team — **Kelowna, BC**
AERODYNAMICS TEAM LEAD — ***May 2025 – Present***

- Designed bike fairing, drafted CAD model, performed FEA & aeroelastic deflection analysis using SolidWorks Suite.
- Led CFD analysis using ANSYS Fluent achieving 15% drag reduction through virtual wind tunnel testing.
- Supervised 5 junior engineering students in CAD & Multi-Physics sims., improving technical proficiency by 40%.

Project "Aegis" - Next Generation Carrier-based Strike Fighter Aircraft Design — ***May 2026***
AIAA Undergraduate Aircraft Design Competition 2026

- Performed MATLAB conceptual sizing, calculating MTOW, fuel fraction, & payload capacity across mission profiles.
- Optimized wing planform, empennage, & control surfaces using XFLR5 for Ma 1.8 cruise & <130 kn landing speed.
- Designed propulsion, landing gear config., weight-&-balance assessment with trade & sensitivity study validation.

Two-Stage Helical Gearbox Design & Analysis for Industrial Power Transmission — ***April 2024***

- Designed compound reverted gearbox achieving 12 HP transmission with 9:1 reduction ratio & bearing selection.
- Performed stress analysis using Goodman criteria / FEA achieving safety factors > 2 for shafts, gears, & bearings.
- Led iterative design reviews, prototype development, & assist in technical documentation; CAD model & part specs.

Advanced Multi-Component Manufacturing & Assembly: Precision-Engineered Scale Train Model — ***December 2023***

- Developed comprehensive CAD & assembly sequences for 15 interconnected components for 0 assembly conflicts.
- Operated 3-axis CNC mills, engine lathes, waterjet cutting, machining tools & conducted QC using RCCA protocols.
- Tested precision assembly of mechanical interfaces, validating fit & function requirements through CMM inspection.

TECHNICAL SKILLS

CAD/CAM	SolidWorks, Rapid Prototyping
CAE/Computational Analysis	ANSYS Fluent CFD, FEA, MATLAB / Simulink, XFLR5
Manufacturing	ANSYS Granta, GD&T, DFM/DFA, ISO / ASTM Standards, Six Sigma Methods
Other	Engineer-in-Training (EIT) Eligibility, Project Management, MS Office Suite

AWARDS

Okanagan Engineering Competition 2025 – 1st Place, Re-Engineering Category — **Kelowna, BC**